

Problem 57. Let $ABCD$ be a cyclic quadrilateral. A point P moves along the arc AD which does not contain B and C . A fixed line ℓ , perpendicular to BC , meets the rays BP , CP at points B_0 , C_0 respectively. Prove that the tangent at P to the circumcircle of triangle PB_0C_0 passes through some fixed point.

Problem 58. Let n be a positive integer. n people take part in a certain party. For any pair of the participants, either the two are acquainted with each other or they are not. What is the maximum possible number of the pairs for which the two are not acquainted but have a common acquaintance among the participants?

Problem 59. Let n be an odd positive integer, and let $x_1, x_2, \dots, x_n$ be non-negative real numbers. Show that

$$\min_{1 \leq i \leq n} (x_i^2 + x_{i+1}^2) \leq \max_{1 \leq j \leq n} (2x_j x_{j+1})$$

where $x_{n+1} = x_1$.

Problem 60. Let ABC be a triangle with $\angle C = 90^\circ$, and let H be the foot of the altitude from C . A point D is chosen inside the triangle CBH so that CH bisects AD . Let P be the intersection point of the lines BD and CH . Let ω be the semicircle with diameter BD that meets the segment CB at an interior point. A line through P is tangent to ω at Q . Prove that the lines CQ and AD meet on ω .